

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from nltk import word_tokenize
from string import punctuation
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import classification_report
from pickle import dump
```

```
data = pd.read_csv("rest_reviews.tsv", sep="\t")
print(data)
```

```
print(data.isnull().sum())
```

```
sw = set(stopwords.words("english"))
sw.remove("not")
sw.remove("don't")
ps = PorterStemmer()
def clean_review(txt):
    txt = txt.lower()
    txt = word_tokenize(txt)
    txt = [t for t in txt if t not in punctuation]
    txt = [t for t in txt if t not in sw]
    txt = [ps.stem(t) for t in txt ]
    txt = " ".join(txt)
    return txt
```

```
data["clean_review"] = data["Review"].apply(clean_review)
print(data)
```



```
tv = TfidfVectorizer()
vector = tv.fit_transform(data["clean_review"])
features = pd.DataFrame(vector.toarray(),
columns=tv.get_feature_names_out())
print(features)
target = data["Liked"]
```

```
x_train, x_test, y_train, y_test = train_test_split(features.values, target)
model = MultinomialNB()
model.fit(x_train, y_train)
```

```
y_pred = model.predict(x_test)
cr = classification_report(y_test, y_pred)
print(cr)
```

```
with open("review_model.pkl", "wb") as f:
    dump(model, f)
    print("model saved")
```

```
with open("review_tv.pkl", "wb") as f:
    dump(tv, f)
    print("tv saved")
```